

BARSAT KHADKA

CONTACT

Hattiesburg, Mississippi, USA
Barsat.Khadka@usm.edu
+1 (210) 739-3050

ONLINE

www.barsat.dev
[LinkedIn](#) · [GitHub](#)

EDUCATION

The University of Southern Mississippi
Hattiesburg, MS
B.S., Computer Engineering GPA: 3.92

2024 – PRESENT

HON THESIS: *Unified Machine Learning in the On-Chip Memory Hierarchy*
B. Khadka, Z. Zhou

RESEARCH INTERESTS

Machine Learning for Chip Design/EDA/Computer Architecture, RL-based World Models, Mechanistic Interpretability

FELLOWSHIPS & PROGRAMS

Break Through Tech AI/ML 2026 – 2027
Selected participant, [Cornell Tech] — applied machine learning fellowship.

San Diego Supercomputer Center CIML JUNE 2026
Summer Institute, UC San Diego — selected for funded HPC/ML training institute.

UR2PhD — Computing Research Association JULY 2026
Undergrad Research to PhD Program.

PUBLICATIONS

CTS-Bench: Benchmarking Graph Coarsening Trade-offs for GNNs in Clock Tree Synthesis

B. Khadka, K. Roxy, M.R. Ahmed

ACCEPTED, MLBench Workshop @ ASPLOS 2026 arxiv.org/pdf/2602.19330

Synthetic Data Generation and Convolutional Neural Network (CNN)-Based Analysis for Precision Agriculture Using Drone Imagery

A. Patrick, M. Khadka, B. Jridi, M.T. Turan, M.S.B. Rashid, M.N. Islam, B. Khadka, S.A. Iftikhar, M.S. Naveed, P. Sengupta

ACCEPTED, SEDE 2026 (35th Int'l Conf. on Software and Data Engineering), San Francisco, CA

TECHNICAL REPORTS

CHIA-OpenROAD: An OpenROAD Extension for Principled, Agentic AI-Driven Hardware/Software Co-Design Research

B. Khadka

A³ Agentic Approaches to Architecture Workshop @ MICRO 2026

PREPRINTS & UNDER REVIEW

ChipDreamer: Learning Chip Physical Design Flow by planning with A World Model

B. Khadka, K. Roxy, M.R. Ahmed

ONGOING

SwiftCTS: Fast Cross-Design Prediction and Pareto Optimization of Clock Tree Metrics via Few-Shot Calibration

B. Khadka, K. Roxy, M.R. Ahmed

RESEARCH
EXPERIENCE

UNDER REVIEW, ACM TDOAES

arxiv.org/pdf/2606.11348

Filter then Verify: A Multiphase Social Engineering Attack Detection Framework using GNN and BERT for email networks

B. Khadka, P. Koirala, K. Neupane, N. Rahimi

UNDER REVIEW, Springer

arxiv.org/pdf/2605.17201

MechRL: Reinforcement Learning Agents Perform Circuit Discovery for Mechanistic Interpretability

B. Khadka (single author)

UNDER REVIEW, IEEE TAI

arxiv.org/pdf/2605.26343

Machine Learning for Chip Design

2025 – PRESENT

Student Researcher

ADVISOR: Dr. Rubel Ahmed, Louisiana Tech (Remote)

- Built CTS-Bench, a benchmark suite evaluating graph-coarsening trade-offs in graph neural networks applied to VLSI clock tree synthesis.
- Designed SwiftCTS, a physics-informed gradient-boosted surrogate (XGBoost/LightGBM) that trains in under 5 s on CPU with sub-millisecond inference; a K-shot calibration scheme cuts power error from 24.5% to 3.3% and wirelength error from 56.6% to under 1% on out-of-distribution macros.
- Led ideation, experimental design, ablation studies, manuscript preparation and ran experiments

Machine Learning for Cybersecurity

AUG. 2025 – PRESENT

Cyber Innovation Lab, USM

ADVISOR: Dr. Nick Rahimi

- Proposed a two-stage “filter-then-verify” framework using inductive Graph Neural Networks (GraphSAGE) and ModernBERT to detect multi-stage social engineering attacks.
- Built a three-phase structural anomaly pipeline (volume, relational, and contextual) to flag anomalous sender–receiver patterns in dynamic email networks.
- Reached 92% precision by adding a co-attention ModernBERT verifier for semantic intent, cutting false positives on the Enron dataset.

Research Software Engineer

NOV. 2025 – PRESENT

EDT (NOAA Research Agent)

Institute for Advanced Analytics & Society, USM

- Built EDT, an “analyst-at-scale” research agent that quantifies and communicates the downstream impact of NOAA oceanographic and environmental data products on public-sector decision-making.
- Engineered a multi-provider model backend (Claude Sonnet 4.6 / GPT-4.5) with Model Context Protocol (MCP) tools querying federal and academic databases, all running inside single-tenant Daytona sandboxes.
- Implemented deterministic skill packs (citation resolution, reference merging, REACH impact scoring, multi-agent research orchestration) so quantitative steps run as real Node.js/Python rather than LLM math.

INDUSTRY EXPERIENCE	Software Engineering Intern (Backend) JUN. 2025 – AUG. 2025 <i>Sports Media Inc (Dakdan Worldwide)</i> Dover, DE - Built Spring Boot microservices and REST APIs to support real-time ad-bidding workflows. - Integrated Apache Kafka to power live inventory updates across 100+ venues. - Contributed to CI/CD pipelines using Jenkins, Maven, and Git; deployed services on AWS and GCP, and wrote JUnit test suites in a Unix/Linux environment.
HONORS & SCHOLARSHIPS	Academic Excellence Award — Full Tuition (4 years) AUG. 2024 Freshman Designation Valedictorian Scholarship AUG. 2024 Honors Scholar Recognition AUG. 2024 1st Runner-Up, VOXO Hackathon NOV. 2024 <i>Reclaiming Time with Generative AI</i> President's List SPRING 2025
SPECIAL PROJECT	Vinaya Journal 2024 – PRESENT <i>Spring Boot, FastAPI, React, SQLite, Chroma, Electron.js</i> Live · GitHub - Built a privacy-first, full-stack desktop journaling app integrating a local LLM + RAG pipeline for offline AI journaling. - Implemented RESTful Spring Boot endpoints with embedded SQLite, plus a FastAPI microservice with Chroma for semantic/vector search, LLM interactions via Ollama, and mood classification. 35+ active users; 150+ GitHub stars; available for Windows and Linux.
TALKS & TUTORIALS	Positive Uses, Negative Uses, and Limitations of Virtual Reality 2024 <i>MAS 89th Annual Conference, Biloxi, MS</i> - Presented findings on VR technology and VRET applications. - Collaborated with peers Anish Kharal and Nhoojah Maharjan in CSC 309; supported by Dr. Sengupta.
TECHNICAL SKILLS	PROGRAMMING: Python, Java, JavaScript/TypeScript, SQL, C++ ML / RESEARCH: PyTorch, HuggingFace Transformers, PEFT, WeightWatcher, LM-Eval-Harness, NumPy, pandas, Chroma FRAMEWORKS: React, Spring Boot, FastAPI, Maven, Tailwind CSS, PostgreSQL TOOLS & DEVOPS: Git, GitHub, Docker, AWS, GCP, Apache Kafka, Jenkins, JUnit, Postman LANGUAGES: English (fluent), Nepali (native), Hindi (advanced)